Nicholas Chandler

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US Citizen

Professional Summary

Data Scientist / ML Engineer with research and production experience across machine learning, LLM systems, and MLOps. Experienced deploying distributed ML pipelines and GPU-accelerated LLM serving on Kubernetes, building agentic AI applications, and developing novel ML methods published in peer-reviewed venues. Strong foundation in statistical modeling, deep learning, and time-series forecasting.

Education

- **Master of Science in Data Science** — Berlin, Germany
  Berlin University of Applied Sciences (BHT) — Graduated With Distinction (1.06 German) — *Oct. 2024 – Sept. 2026*
- **Bachelor of Science in Computer Science; Minors in Math & Stats** — Bellingham, WA
  Western Washington University — GPA: 3.96 — *Sept. 2020 – June 2024*

Experience

- **Data Scientist / ML Researcher** — Berlin, Germany
  Calgo Lab — Berlin University of Applied Sciences (BHT) — Studentische Hilfskraft — *Feb. 2025 – Sept. 2026*
  - Engineered distributed ML experiment pipelines on a Kubernetes cluster with Helm; automated testing and linting via GitHub Actions CI/CD.
  - Deployed and maintained Dockerized LLM serving stack (vLLM on NVIDIA GPUs) and web-based research demos, including a demo used by the Charité research hospital in Berlin.
  - Architected **MechDetect**, a novel ML framework for classifying error mechanisms in tabular datasets; published at IEEE DSIS 2025.
  - Implemented spatiotemporal groundwater-level forecasting (master's thesis) combining Chronos2, N-HiTS, and GPyTorch Gaussian Processes.
- **Machine Learning Researcher** — Bellingham, WA
  Hutch Research — Western Washington University — *Mar. 2023 – June 2024*
  - Built a PyTorch/TensorFlow probabilistic neural network pipeline to predict binary star system parameters; published in *The Astronomical Journal*, Vol. 171 (2026).
  - Optimized large astronomical dataset processing with NumPy; maintained code quality and reproducibility in a 4-person team.
- **Statistics Researcher** — Bellingham, WA
  Western Washington University — *Oct. 2022 – June 2024*
  - Developed a novel class of non-parametric tests in R; presented at Joint Mathematics Meetings 2024 and SIAM PNW 2023.
  - Created scientific visualizations and ran distributed computing workflows for experiments on an on-prem compute cluster.

Selected Projects

- **LangGraph Email Agent** | *vLLM, LangGraph, Kubernetes, Docker, Nvidia A100*
  - Deployed an agentic LLM email-answering system on Kubernetes using vLLM for inference serving and LangGraph for multi-step orchestration with structured output.
- **Tax Law RAG System** | *BERT Fine-tuning, LangChain, FAISS, Kubernetes*
  - Built a production RAG pipeline with fine-tuned BERT retrieval; benchmarked FAISS vs. Annoy vector databases on English and German corpora. Published the fine-tuned BERT to HuggingFace.
- **Berlin PM2.5 Spatiotemporal Modeling** | *Chronos2, Kubernetes, Docker, Streamlit*
  - Automated daily ETL and inference pipeline on Kubernetes; spatiotemporal forecasting via Chronos2 foundation model and gaussian processes.
- **tab-err: Tabular Data Error Library** | *Python, PyPI, CI/CD, GitHub Actions*
  - Contributed to open-source library for realistic tabular error generation; developed high-level APIs and unit tests.

Publications

- **Anonymized but Useful Synthetic Tabular Health Data for AI based Fall Risk Assessment** | *Scientific Data* 2026
- **Probabilistic Neural Networks for Eclipsing Binary Parameters** | *The Astronomical Journal* 2026
- **MechDetect: Detecting Data-Dependent Errors** | IEEE DSIS 2025
- **Towards Realistic Error Models for Tabular Data** | *ACM JDIQ* 2025

Skills

MLOps & Deployment: Kubernetes, Docker, Helm, vLLM, FastAPI, GitHub Actions, Prometheus, Grafana
LLM & Agentic AI: LangChain, LangGraph, FAISS, RAG, Prompt Engineering, Structured Outputs, Agentic Pipelines, Transformer Fine-tuning
ML & Data: PyTorch, TensorFlow, GPyTorch, Scikit-learn, Pandas, Numpy, Matplotlib, MySQL, Time-Series, Spatial Data
Cloud & DevOps: AWS, GPU Workloads (NVIDIA), Containerization, Git, Linux
Languages: Python, SQL, R, Go, Bash, Java **Spoken:** English (Native), German (B1)